

MAE 263F – Homework 2

Yuchen Wang (Ph.D. in Mechanical Engineering)

University of California, Los Angeles

October 22, 2025

I. TIME-DEPENDENT MAXIMUM DEFLECTION

Question: Plot the maximum vertical displacement, $y_{\max}$, of the beam as a function of time. Depending on your coordinate system, $y_{\max}$ may be negative. Does $y_{\max}$ eventually reach a steady value? Examine the accuracy of your simulation against the theoretical prediction from Euler beam theory:

$$y_{\max} = \frac{Pc(l^2 - c^2)^{1.5}}{9\sqrt{3}EI} \quad \text{where} \quad c = \min(d, l - d) \quad (1)$$

Answer: The maximum vertical displacement, $y_{\max}$, of the beam as a function of time as shown in Fig 1.

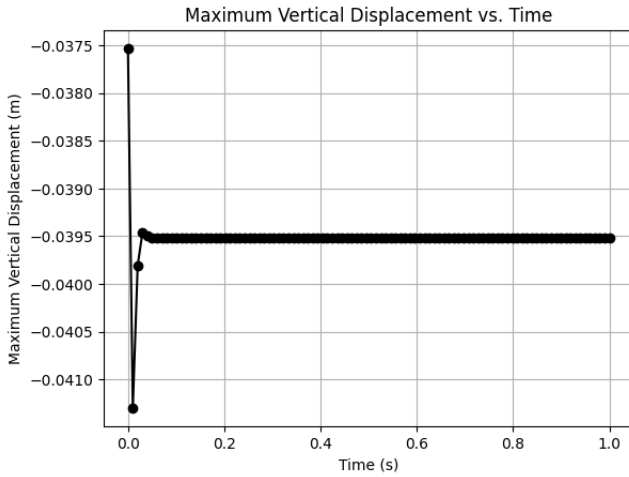


Fig. 1. The maximum vertical displacement $y_{\max}$ of the beam as a function of time under static loading

From the plot in Fig.1, the response shows a short transient (initial overshoot) and then settles to a constant value by approximately $t \approx 0.02$ s. The steady-state deflection is

$$y_{\max} \approx -0.0395 \text{ m,}$$

where the negative sign indicates downward displacement.

Compared with Euler–Bernoulli beam theory, the theoretical maximum displacement is $|y_{\max}|_{\text{theory}} = 0.038045$ m, while the simulated result gives $|y_{\max}|_{\text{simulation}} = 0.039519$ m. The percentage difference between them is 3.88%. This small difference is expected and can be attributed to factors such as:

- geometric nonlinearities or large rotations neglected in the Euler–Bernoulli formulation,
- discretization errors in the numerical mesh and time integration,
- boundary condition and load application approximations.

II. SIMULATION ADVANTAGE AT LARGE DEFORMATION

Question: What is the benefit of your simulation over the predictions from beam theory? To address this, consider a higher load $P = 20,000$ N such that the beam undergoes large deformation. Compare the simulated result against the prediction from beam theory. Euler beam theory is only valid for small deformation whereas your simulation, if done correctly, should be able to handle large deformation. You should create a plot of P ($20 \text{ N} \leq P \leq 20,000 \text{ N}$) vs. $y_{\max}$ using data from both simulation and beam theory and quantify the value of P where the two solutions begin to diverge.

Answer: Euler–Bernoulli beam theory assumes (i) small deflections/rotations, (ii) a straight neutral axis, and (iii) no shear deformation or material nonlinearity. A properly set-up numerical simulation can relax these assumptions: it can capture geometric nonlinearity (large rotations and mid-plane stretching), shear deformation, nonlinear/inhomogeneous boundary loading, and, if enabled, material nonlinearity and contact. Consequently, the simulation remains valid as the load increases and the beam experiences large deformation, whereas the linear theory does not.

In the case of $P = 20,000$ N, the theoretical maximum displacement is $|y_{\max}|_{\text{theory}} = 0.380449$ m, while the simulated result gives $|y_{\max}|_{\text{simulation}} = 0.245459$ m. The percentage difference between them is 35.48%. This significant deviation indicates that the Euler–Bernoulli beam theory is no longer valid under such a high load, since it assumes small deflection and linear geometry. In contrast, the numerical simulation accounts for geometric nonlinearity, allowing it to capture large rotations and mid-plane stretching effects that reduce the effective stiffness of the beam. Consequently, the simulated displacement is smaller than the theoretical prediction, reflecting the stiffening behavior that occurs when the beam geometry changes substantially during deformation.

The results for loads in the range $20 \text{ N} \leq P \leq 20,020 \text{ N}$ with an increment of 2,500 N are summarized in Table 1, showing the theoretical values of $y_{\max}$, the simulated results,

and the corresponding percentage differences at each load step. Figure 2 presents a comparison between the simulation results and the Euler–Bernoulli beam theory predictions. If we assume that the acceptable deviation should not exceed 5%, then the divergence between the two solutions begins in the range of $P = 5,020\text{ N}$ to $7,520\text{ N}$.

TABLE I
COMPARISON OF THEORETICAL AND SIMULATED MAXIMUM VERTICAL
DISPLACEMENTS $y_{\max}$ UNDER DIFFERENT POINT LOADS P

P (N)	$y_{\max}^{\text{theory}}$ (m)	$y_{\max}^{\text{sim}}$ (m)	Diff. (%)
20	0.000380	0.000398	4.67
2520	0.047937	0.049530	3.32
5020	0.095493	0.094774	0.75
7520	0.143049	0.133848	6.43
10020	0.190605	0.166267	12.77
12520	0.238161	0.192652	19.11
15020	0.285717	0.214014	25.10
17520	0.333273	0.231365	30.58
20020	0.380830	0.245562	35.52

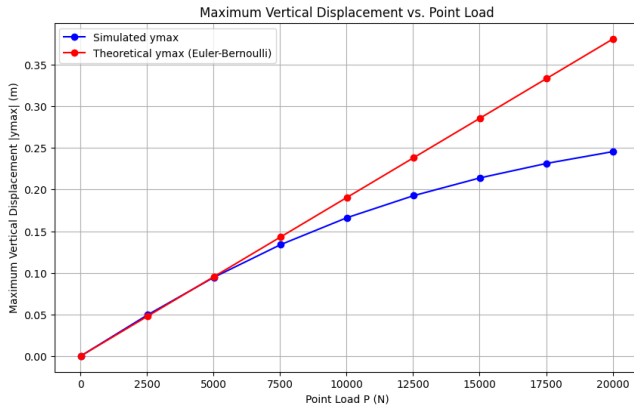


Fig. 2. Comparison of the maximum vertical displacement $y_{\max}$ obtained from simulation and Euler–Bernoulli beam theory

For small P , the two curves overlap (linear regime). As P grows, $y_{\max}^{\text{sim}}$ exceeds $y_{\max}^{\text{theory}}$ because geometric nonlinearity effectively softens the response (the tangent stiffness drops as rotations increase).

CODE AND SOURCE ATTRIBUTION

The source code is adapted from *Simulation of a Beam with Fixed DOFs* and has been refined and formatted with the assistance of ChatGPT.